

PRATYUSH PADHY

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EDUCATION

University of California, Irvine

Irvine, CA

B.S., Computer Science, Specialization in Intelligent Systems — GPA: 3.86/4.00

Expected Graduation: 2028

• Dean's Honor List: Fall '25, Winter '26, Spring '26 | **Organizations:** Data@UCI, AI@UCI, Cyber@UCI, Hack@UCI

PROFESSIONAL EXPERIENCE

Tata Consultancy Services (TCS)

CMU Campus, Pittsburgh, PA

AI/ML Intern — TCS Medical Robotics Center

Jun. 2026 – Present

- Architected a **5-stage behavior tree** pipeline for autonomous manipulation (**ROS2 Jazzy, Gazebo Harmonic, py_trees**) with automated replanning across **6** recovery strategies, validated by **7** passing unit tests.
- Trained a **PPO** policy (**Stable-Baselines3, Gymnasium**) in **7 minutes** on **CPU-only** hardware in a fast kinematic environment, validating forward kinematics to millimeter accuracy against the physics simulator for **99%** target-reach success (**<25mm**) with zero-shot transfer to the full simulator under real-world conditions.
- Fine-tuned a **450M-parameter** VLA model (**SmolVLA, LeRobot**) via **imitation learning** on **40** demos for CPU-only inference (**4GB** RAM), benchmarked against PPO: **70%** success at 10-20s/trial vs. **33%** at 3min/trial.

Ready Tutor

Irvine, CA

Software Developer Intern

Mar. 2026 – Jun. 2026

- Engineered a scalable **MongoDB/SQLite** schema class with **React** useContext integration, replacing **10+** component-level instances and refactoring **5+** database files to centralize and unify storage across application stack.
- Built a Fall Quarter course catalog with real-time retrieval of **1,200+** UCI courses, enabling students to browse and plan schedules; resolved **30+** bugs including a credits calculation issue, contributing to securing **\$5,500** in funding.
- Redesigned the year planning data model with a **JSON-stringified** array schema, replacing a rigid 4-year structure to support fully dynamic multi-year academic scheduling across fall, winter, spring, and summer quarters.

Robotics for All

Palo Alto, CA

Software Intern

Jun. 2023 – Aug. 2024

- Deployed a comprehensive asynchronous training system with structured readiness assessments for **24+** instructors, standardizing the onboarding pipeline and ensuring consistent teaching quality across all program locations.
- Coached coding classes of **20** students and designed a parent feedback system to continuously improve curriculum quality; interviewed volunteers to assess readiness; awarded the Presidential Service Award (Gold) for **250+** hours.

PROJECTS

Chronicle | Python, FastAPI, PostgreSQL, Next.js, TypeScript | chronicles-weld.vercel.app

Jun. 2026 – Present

- Built a job aggregation platform ingesting live postings from **200+** companies' ATS systems (**Greenhouse, Lever, Ashby**) via async fan-out with per-company fault isolation, then normalizing and deduplicating cross-source postings.
- Engineered a hybrid semantic search and recommendation layer (**pgvector** HNSW, int8 **ONNX** embeddings, **RRF** fusion) at **p95 <35ms** on 10K+ jobs, lifting **NDCG@10** 0.853→0.943 and **MRR** to 1.000 over a rule-based baseline.
- Shipped **Google OAuth** accounts, kanban application tracking, saved-search email alerts, and a **Manifest V3** browser extension that autofills **Greenhouse/Lever/Ashby** applications from a stored profile.

FacePulse | Python, TensorFlow, FastAPI, React, OpenCV, Dlib | face-pulse.netlify.app

Jan. 2026 – Jun. 2026

- **Featured on Handshake AI Showcase**; evolved from a facial landmark detector into a comparative study of **MLP, CNN, and VGG16** on **FER2013** (**35,000+** images) via transfer learning and evaluation.
- Built a full-stack web app with a **FastAPI + TensorFlow** backend deployed via **Docker** on **Hugging Face Spaces** and a **React/TypeScript** frontend on Netlify with real-time webcam emotion detection and live probability outputs.
- Implemented an analytics dashboard tracking real-time emotion probability charts, per-session timelines, cumulative mood trends, and emotion detection with smooth animated transitions using **Chart.js** and **Framer Motion**.

Argus-AI | Python, YOLOv8, FastAPI, Next.js, TypeScript, Docker | ai-argus.netlify.app

Apr. 2026 – Jun. 2026

- Trained a custom **YOLOv8s** model on **5,200+** dashcam images across **23** object classes, enabling real-time detection of red light running, stop sign breaches, no-entry violations, and pedestrian proximity risks across urban environments.
- Designed a spatial rule engine that infers traffic violations by analyzing proximity relationships between detected bounding boxes, flagging vehicles crossing stop lines, entering no-entry zones, and pedestrian encroachments.
- Deployed a full-stack application with a **FastAPI + Python** backend containerized via **Docker** and hosted on **Google Cloud Run**, paired with a **Next.js/TypeScript** frontend on Netlify with annotated violation reports.

TECHNICAL SKILLS

Languages: Python, Java, R, SQL, HTML, CSS, JavaScript, TypeScript

Libraries & Frameworks: PyTorch, TensorFlow, scikit-learn, Transformers, OpenCV, NumPy, Pandas, Matplotlib, YOLOv8, Stable-Baselines3, Gymnasium, React, Next.js, FastAPI, Tailwind CSS, Chart.js, LLMs, RAG, NLP, Reciprocal Rank Fusion, recommender systems, REST APIs

Tools & Platforms: Git, Docker, pytest, PostgreSQL, pgvector, HNSW, ROS2, Gazebo, Vercel, Render, Google Cloud Run, Hugging Face Spaces, MongoDB, SQLite